1 B3 data

Adjusted prices → returns

58
equities

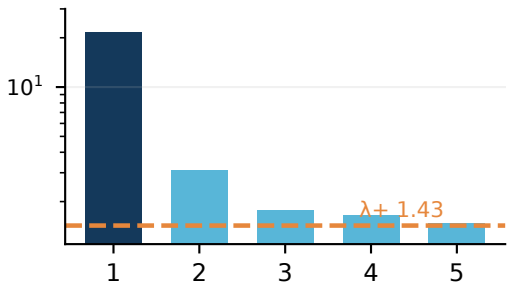
1,527
common dates

23 Aug 2006 – 15 Aug 2024

Complete synchronized panel

2 RMT modes

Empirical eigenvalues (log scale)



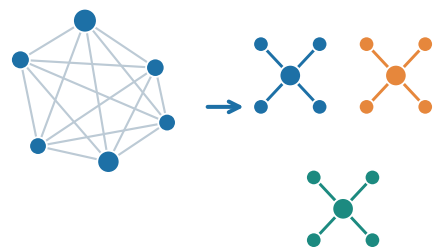
1 market + 4 additional modes

2 additional modes after issuer collapse

Same 1,527 dates

3 Network structure

Original **Group mode**

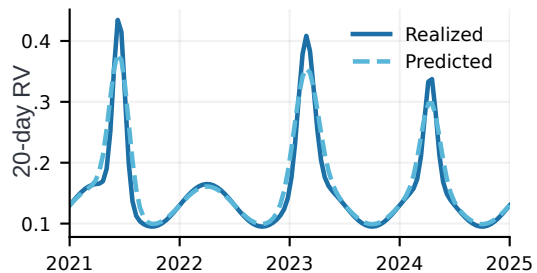


Schematic network views

Weaker links; localized organization remains

4 Forecast test

Illustrative 20-day realized volatility



Illustrative only

Held-out test: 2021–2025

RV is not annualized

KEY FINDING

Adding network features did not improve forecasts: Ridge models using them had higher test loss (QLIKE) than the market-only Ridge model.